Sequences — problem 9.24

A. Engel — Problem-Solving Strategies · 1998

Three sequences x_n, y_n, z_n with positive initial terms x_1, y_1, z_1 are defined for $n \geq 1$ by $x_{n+1} = y_n + 1/z_n$, $y_{n+1} = z_n + 1/x_n$, $z_{n+1} = x_n + 1/y_n$. Show that

(a) none of the three sequences is bounded.

(b) At least one of $x_{200}, y_{200}, z_{200}$ is greater than 20.